

SYRACUSE HANCOCK, NY, USA

WMO: 725190

Lat: 43.111N

Lon: 76.104W

Elev: 413

StdP: 14.48

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14771

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.5	4.1	-8.7	3.5	0.0	-3.7	4.6	5.7	30.0	30.3	26.9	24.2	6.4	90	0.540

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	89.1	73.1	86.4	71.4	83.8	69.9	75.4	85.1	73.8	82.6	72.2	80.6	10.2	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.5	122.1	80.5	70.8	115.1	78.7	69.2	109.1	77.4	39.0	85.5	37.5	82.6	36.1	80.5	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.4	20.7	18.5	-9.1	93.5	7.2	3.4	-14.3	95.9	-18.6	97.9	-22.6	99.8	-27.9	102.2		
			-9.6	78.5	7.1	1.9	-14.7	79.8	-18.9	80.9	-22.9	82.0	-28.1	83.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.9	24.1	26.0	34.0	46.5	58.5	67.4	72.1	70.6	63.5	51.9	40.7	30.7
	DBStd	19.01	12.31	10.60	10.63	9.10	8.27	6.72	5.43	5.29	7.15	8.01	8.88	9.50
	HDD50	3149	806	674	508	170	16	0	0	0	1	74	301	599
	HDD65	6504	1269	1093	961	558	238	52	5	12	113	413	728	1062
	CDD50	2765	2	1	12	66	279	521	685	638	405	132	22	2
	CDD65	644	0	0	0	4	36	123	225	185	66	5	0	0
	CDH74	5482	0	0	12	76	387	1089	1949	1393	532	44	0	0
	CDH80	1608	0	0	1	18	99	331	616	396	143	4	0	0
Wind	WSAvg	8.2	9.6	9.6	9.4	9.2	7.8	7.1	6.7	6.3	6.7	7.8	8.8	9.3
Precipitation	PrecAvg	37.7	2.5	2.1	3.0	3.5	3.4	3.2	3.5	3.4	3.5	3.0	3.5	3.0
	PrecMax	52.8	6.3	4.5	5.4	7.7	7.8	10.3	8.0	5.9	7.4	6.7	5.8	5.2
	PrecMin	27.8	1.1	0.6	1.0	1.4	0.6	1.0	0.7	1.0	0.7	0.2	1.3	1.4
	PrecStd	5.9	1.1	0.9	1.1	1.5	1.7	1.7	1.8	1.3	1.5	1.5	1.1	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.3	60.6	73.6	82.9	87.3	91.1	92.6	92.1	89.2	79.2	69.3	62.6
		MCWB	55.7	52.2	58.9	62.9	69.2	74.6	75.3	74.1	72.9	66.1	58.8	54.1
	2%	DB	52.6	51.2	62.4	74.5	82.8	87.5	89.2	87.4	84.0	73.8	64.3	53.4
		MCWB	48.1	44.6	51.8	58.1	67.0	71.7	74.2	72.0	70.3	63.1	55.0	47.9
	5%	DB	45.6	45.5	56.2	68.1	78.4	83.9	86.2	84.3	80.1	69.8	60.4	48.4
		MCWB	41.9	40.1	47.7	54.6	64.5	69.5	71.9	70.5	68.7	61.3	53.0	44.3
	10%	DB	40.3	40.4	50.1	62.8	73.8	80.4	83.4	81.5	76.2	65.3	55.0	43.5
		MCWB	36.7	36.2	43.2	52.1	61.5	68.0	70.3	69.4	66.6	57.8	49.1	39.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.5	53.4	60.2	65.9	72.7	76.4	78.5	77.0	75.3	69.1	60.4	55.5
		MCDB	60.8	58.8	71.1	78.9	84.1	87.7	89.2	86.8	83.6	73.8	66.7	61.6
	2%	WB	48.0	45.3	53.6	60.5	69.0	73.9	75.8	74.6	72.6	65.3	57.2	49.0
		MCDB	51.9	50.4	60.9	70.8	79.3	84.0	85.7	82.8	81.3	72.2	62.5	52.3
	5%	WB	42.0	40.5	48.5	56.9	66.4	71.8	74.0	72.8	70.1	62.0	53.4	44.4
		MCDB	45.2	45.0	55.4	66.0	75.6	80.7	82.9	80.7	77.6	68.5	59.1	48.4
	10%	WB	37.0	36.4	43.6	52.7	63.8	69.7	72.3	71.1	67.9	58.8	49.6	40.0
		MCDB	39.9	40.1	50.2	61.1	71.7	77.4	80.5	78.8	74.4	64.6	54.6	43.2
Mean Daily Temperature Range	5% DB	MDBR	14.5	15.1	16.3	19.4	20.6	19.2	18.8	18.8	19.3	16.7	14.6	12.5
		MCDBR	19.7	22.1	26.0	30.1	26.3	23.7	21.8	22.1	23.7	23.5	22.6	18.9
		MCWBR	16.7	17.3	16.9	17.0	13.6	11.5	10.0	10.0	11.8	14.3	15.8	15.4
	5% WB	MCDBR	19.1	20.6	24.5	26.6	23.2	20.1	18.8	18.5	20.4	20.6	20.5	17.9
		MCWBR	16.6	17.0	17.4	16.8	13.1	11.3	9.9	9.3	10.9	13.5	15.6	15.4
Clear-Sky Solar Irradiance	taub	0.296	0.310	0.335	0.364	0.393	0.437	0.445	0.429	0.377	0.356	0.315	0.291	
	taud	2.355	2.330	2.385	2.367	2.329	2.231	2.215	2.245	2.377	2.435	2.502	2.455	
	Ebn at Noon	265	281	288	286	279	266	262	262	270	261	259	259	
	Edh at Noon	25	31	34	37	39	44	44	41	34	28	22	21	
All-Sky Solar Radiation	RadAvg	417	681	1063	1395	1693	1796	1839	1636	1302	785	495	322	
	RadStd	44	82	92	162	168	156	153	87	121	82	63	46	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page